

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398014
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Herrmann; Douglas K.

Multidirectionally stepped auto-collator system and method

Abstract

An automated media sheet processing system for automated collating, stacking and transferring media sheets exiting a media source includes a dynamically oriented baffle having a plurality of stepped bins. Each stepped bin is slidably arranged adjacent another stepped bin, with the stepped bins configured to receive media sheets and allow them to fall onto and accumulate on the stepped bins. The stepped bins may move and change orientation into a stair configuration descending in different directions. An automated reciprocating pusher may move in the different directions orthogonally to the stepped bins to move and collate the media sheets into collated stacks downstream the stepped bins on different or opposite sides of the dynamically oriented baffle, with no reset or extraneous movement needed between reciprocating pushes. No reset is needed because the reciprocating pusher ends a first push in position to start a next push to the opposite side.

Inventors:	Herrmann; Douglas K. (Webster, NY)
Applicant:	Xerox Corporation (Norwalk, CT)
Family ID:	1000008780904
Assignee:	Xerox Corporation (Norwalk, CT)
Appl. No.:	18/536815
Filed:	December 12, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250187865 A1	Jun. 12, 2025

Publication Classification

Int. Cl.: B65H39/02 (20060101); B65H39/115 (20060101)

U.S. Cl.:

CPC **B65H39/115** (20130101); B65H2301/433 (20130101); B65H2301/43824 (20130101); B65H2404/733 (20130101)

Field of Classification Search

CPC: B41L (43/10); B41F (13/64); B65H (39/02); B65H (39/075); B65H (39/055); B65H (39/115); B65H (2301/433); B65H (2301/435); B65H (2301/43824); B65H (2404/73); B65H (2404/733)

USPC: 270/52.16; 270/52.29; 270/58.29

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9463945	12/2015	Herrmann	N/A	N/A
9463946	12/2015	Herrmann	N/A	N/A
9527693	12/2015	Herrmann	N/A	N/A
9592989	12/2016	Herrmann	N/A	B65H 31/3009
9624062	12/2016	Herrmann	N/A	B65H 31/34
10071877	12/2017	Herrmann	N/A	B65H 31/02
10525509	12/2019	Chapelet	N/A	B65G 47/46
10968068	12/2020	Herrmann et al.	N/A	N/A
11584607	12/2022	Ludgate et al.	N/A	N/A
12116156	12/2023	Herrmann	N/A	B65H 31/36
2016/0185557	12/2015	Herrmann	N/A	N/A
2017/0129733	12/2016	Herrmann	N/A	N/A

Primary Examiner: Nicholson, III; Leslie A

Attorney, Agent or Firm: Caesar Rivise, PC

Background/Summary

FIELD OF DISCLOSURE

(1) This invention relates generally to an automated collation system, and more particularly, to a reversing stepped auto-collation system that automatically collects and collates retail signage destined for in-store shelves.

BACKGROUND

(2) Retail stores often utilize signage (e.g., edge markers) to convey information regarding products offered for sale, e.g., product costs, unit cost, sale pricing, etc. Such markers must be updated and/or replaced on a periodic basis. For example, regular product pricing may change, or during a sale, a discounted price may be necessary. Changes to edge markers may be required for hundreds or even thousands of products and these changes may be required daily weekly or another periodic term. In addition, product placement may change which would require updating of the edge markers. In some states, it is critical that the edge markers be updated in a timely fashion as the

retail store may be obligated to honor the price displayed adjacent the product. In other words, if the store fails to remove the edge marker that displays a discounted cost, the store must charge that cost if a customer relies upon that price when making a purchase selection. In view of the foregoing, it should be apparent that proper timing and placement of edge markers is a critical responsibility of a retail store.

(3) Although some retail chain stores share common store layouts, also known as a store planogram, most retail locations, even within a chain store have unique store planograms. The changeover of store signage can incur significant time which in turn incurs significant cost. A common practice is to print sheets of edge marker strips and an employee or group of employees are tasked with edge marker changeover. These methods include various deficiencies, e.g. edge marker strips compiled out of order or not matched to the store planogram, sheets that require further separation of individual store departments, etc. These methods are quite costly and presently, in at least one instance, requires for example, 20 people employed to individually catch and collate each sheet of edge markers. Other media collating systems including U.S. Pat. Nos. 9,463,945 B2, 9,463,946 B2 and 9,527,693 B2, are known, but the heretofore-mentioned problems persist.

(4) Obviously, there is a need for a more efficient shelf edge marker collation system that presents shelf edge markers to store employees in a per store planogram order for in-store deployment. U.S. Pat. No. 10,968,068 B1 discloses such a system. Still it would be more beneficial to provide an automated solution for collating and processing the edge marker strips with greater efficiencies for subsequent shipping to a store.

SUMMARY

(5) The following presents a simplified summary in order to provide a basic understanding of some aspects of one or more embodiments or examples of the present teachings. This summary is not an extensive overview, nor is it intended to identify key or critical elements of the present teachings, nor to delineate the scope of the disclosure. Rather, its primary purpose is merely to present one or more concepts in simplified form as a prelude to the detailed description presented later. Additional goals and advantages will become more evident in the description of the figures, the detailed description of the disclosure, and the claims.

(6) The foregoing and/or other aspects and utilities embodied in the present disclosure may be achieved by providing a media sheet processing apparatus for automatically collating and transferring media sheets exiting an upstream source. The media sheet processing apparatus includes a dynamically oriented baffle and an automated reciprocating pusher. The dynamically oriented baffle has a plurality of stepped bins, each stepped bin slidably arranged adjacent another one of the plurality of stepped bins and having a longitudinal surface to receive a first set of the media sheets deposited thereon. The dynamically oriented baffle is configured to first shift the plurality of stepped bins vertically and change height with respect to an adjacent stepped bin, the first shift of the stepped bins resulting in the plurality of stepped bins having a first stair configuration descending in a first direction. The automated reciprocating pusher is configured for automatic bidirectional movement in a cross process direction to the deposited media sheets. Further, the automated reciprocating pusher is configured during a first operation to move the first set of media sheets in the first direction from the stepped bins in the first stair configuration into a first compiled collated stack at a first position, and during a second operation to move an additional set of media sheets in a second direction opposite the first direction. The dynamically oriented baffle is further configured to dynamically second shift the stepped bins vertically and change vertical orientation with respect to the adjacent stepped bin, with the second shift of the stepped bins resulting in the plurality of stepped bins having a second stair configuration descending in the second direction.

(7) According to aspects illustrated herein, an exemplary media sheet processing method for automatically collating and transferring media sheets exiting an upstream source includes providing

a dynamically oriented baffle having a plurality of stepped bins, each stepped bin slidably arranged adjacent another one of the plurality of stepped bins and having a longitudinal surface to receive media sheets deposited thereon, the dynamically oriented baffle configured to first shift the plurality of stepped bins vertically and change height with respect to an adjacent stepped bin, the first shift of the stepped bins resulting in the plurality of stepped bins having a first stair configuration descending in a first direction; receiving a first set of the media sheets from the upstream source onto the longitudinal surfaces of the plurality of stepped bins; providing an automated reciprocating pusher configured for automatic bidirectional movement in a cross process direction to the deposited media sheets; during a first operation, moving the automated reciprocating pusher in a first direction to move the first set of the media sheets from the stepped bins in the first stair configuration into a first compiled collated stack at a first position; and dynamically shifting the stepped bins vertically and into a different vertical orientation with respect to the adjacent stepped bin resulting in the plurality of stepped bins having a second stair configuration descending in a second direction opposite the first direction, the plurality of stepped bins further configured to receive a second set of media sheets deposited thereon. The method may also include, during a second operation, moving the automated reciprocating pusher in a second direction opposite the first direction to move the media sheets from the stepped bins in the second stair configuration into a second compiled collated stack at a second position distanced from the first position.

(8) According to aspects described herein, an exemplary media sheet processing system for automatically collating and transferring media sheets exiting an upstream source includes a dynamically oriented baffle, a compiler, and an automated reciprocating pusher. The dynamically oriented baffle has a plurality of stepped bins, each stepped bin slidably arranged adjacent another one of the plurality of stepped bins and having a longitudinal surface to receive a first set of the media sheets deposited thereon. The dynamically oriented baffle is configured to first shift the plurality of stepped bins vertically and change height with respect to an adjacent stepped bin, the first shift of the stepped bins resulting in the plurality of stepped bins having a first stair configuration descending in a first direction. The compiler is positioned along the processing path between the upstream source and the dynamically oriented baffle, with the compiler being operatively connected to the dynamically oriented baffle. The compiler is configured to receive the media sheets exiting the upstream source and to temporarily hold the media sheets, and following one of the first operation and the second operation, the compiler is further configured to release the held media sheets to the dynamically oriented baffle. The automated reciprocating pusher is configured for automatic bidirectional movement in a cross process direction to the deposited media sheets, wherein, in response to detecting that the first set of the media sheets have settled into the bins for collation thereof, the automated reciprocating pusher is configured during a first operation to move the first set of the media sheets in the first direction from the stepped bins in the first stair configuration into a first compiled collated stack at a first position.

(9) The dynamically oriented baffle is further configured to dynamically second shift the stepped bins vertically and change vertical orientation with respect to the adjacent stepped bin, the second shift of the stepped bins resulting in the plurality of stepped bins having a second stair configuration descending in a second direction opposite the first direction. The longitudinal surfaces of the stepped bins are further configured to receive a second set of the media sheets deposited thereon. In response to detecting that a second set of media sheets have settled into the bins for collation thereof, the automated reciprocating pusher is further configured during a second operation to move the second set of the media sheets in the second direction from the stepped bins in the second stair configuration into a second compiled collated stack at a second position distanced from the first position. The automated reciprocating pusher is further configured to move bidirectionally horizontally in the first direction and second direction without vertical movement therebetween.

(10) Exemplary embodiments are described herein. It is envisioned, however, that any system that incorporates features of apparatus and systems described herein are encompassed by the scope and spirit of the exemplary embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various exemplary embodiments of the disclosed apparatuses, mechanisms and methods will be described, in detail, with reference to the following drawings, in which like referenced numerals designate similar or identical elements, and:
- (2) FIG. 1 is a schematic diagram of an exemplary media sheet processing system according to systems, apparatuses and methods herein;
- (3) FIG. 2 is a partial perspective view of a related art media sheet processing apparatus media path;
- (4) FIG. 3 is a perspective view schematic diagram of a related art media sheet processing apparatus;
- (5) FIG. 4 is a side view of a related art stacking/collating device showing a series of angled baffles;
- (6) FIG. 5 is a side view of a collating system in accordance with examples of the embodiments;
- (7) FIG. 6 is a side view of the collating system of FIG. 5 according to systems and methods herein;
- (8) FIG. 7 is a side view of the collating system of FIG. 5 according to systems and methods herein;
- (9) FIG. 8 is a side view of the collating system of FIG. 5 according to systems and methods herein;
- (10) FIG. 9 is a top view of a media sheet processing system in accordance with examples of the embodiments;
- (11) FIG. 10 is a flowchart depicting the operation of an exemplary automated media sheet processing system; and
- (12) FIG. 11 is a block diagram of a controller for executing instructions to control the automated media sheet processing system.

DETAILED DESCRIPTION

(13) Illustrative examples of the devices, systems, and methods disclosed herein are provided below. An embodiment of the devices, systems, and methods may include any one or more, and any combination of, the examples described below. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth below. Rather, these exemplary embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. Accordingly, the exemplary embodiments are intended to cover all alternatives, modifications, and equivalents as may be included within the spirit and scope of the apparatuses, mechanisms and methods as described herein.

(14) We initially point out that description of well-known starting materials, processing techniques, components, equipment and other well-known details may merely be summarized or are omitted so as not to unnecessarily obscure the details of the present disclosure. Thus, where details are otherwise well known, we leave it to the application of the present disclosure to suggest or dictate choices relating to those details. The drawings depict various examples related to embodiments of illustrative methods, apparatuses, and systems for automatically collecting, collating and transporting media sheets (e.g., workpieces, retail edge marker strips) destined for in-store shelves.

(15) When referring to any numerical range of values herein, such ranges are understood to include

each and every number and/or fraction between the stated range minimum and maximum. For example, a range of 0.5-6% would expressly include the endpoints 0.5% and 6%, plus all intermediate values of 0.6%, 0.7%, and 0.9%, all the way up to and including 5.95%, 5.97%, and 5.99%. The same applies to each other numerical property and/or elemental range set forth herein, unless the context clearly dictates otherwise.

(16) The modifier “about” used in connection with a quantity is inclusive of the stated value and has the meaning dictated by the context (for example, it includes at least the degree of error associated with the measurement of the particular quantity). When used with a specific value, it should also be considered as disclosing that value. For example, the term “about 2” also discloses the value “2” and the range “from about 2 to about 4” also discloses the range “from 2 to 4.”

(17) The term “controller” or “control system” is used herein generally to describe various apparatus such as a computing device relating to the operation of one or more device that directs or regulates a process or machine. A controller can be implemented in numerous ways (e.g., such as with dedicated hardware) to perform various functions discussed herein. A “processor” is one example of a controller which employs one or more microprocessors that may be programmed using software (e.g., microcode) to perform various functions discussed herein. A controller may be implemented with or without employing a processor, and also may be implemented as a combination of dedicated hardware to perform some functions and a processor (e.g., one or more programmed microprocessors and associated circuitry) to perform other functions. Examples of controller components that may be employed in various embodiments of the present disclosure include, but are not limited to, conventional microprocessors, application specific integrated circuits (ASICs), and field-programmable gate arrays (FPGAs).

(18) Embodiments as disclosed herein may also include computer-readable media for carrying or having computer-executable instructions or data structures stored thereon. Such computer-readable media can be any available media that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, such computer-readable media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code means in the form of computer-executable instructions or data structures. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or combination thereof) to a computer, the computer properly views the connection as a computer-readable medium. Thus, any such connection is properly termed a computer-readable medium. Combinations of the above should also be included within the scope of the computer-readable media.

(19) Computer-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions. Computer-executable instructions also include program modules that are executed by computers in stand-alone or network environments. Generally, program modules include routines, programs, objects, components, and data structures, and the like that perform particular tasks or implement particular abstract data types. Computer-executable instructions, associated data structures, and program modules represent examples of the program code means for executing steps of the methods disclosed herein. The particular sequence of such executable instructions or associated data structures represents examples of corresponding acts for implementing the functions described therein.

(20) Although embodiments of the invention are not limited in this regard, discussions utilizing terms such as, for example, “processing,” “computing,” “calculating,” “determining,” “using,” “establishing,” “analyzing,” “checking,” or the like, may refer to operation(s) and/or process(es) of a controller, computer, computing platform, computing system, or other electronic computing device, that manipulate and/or transform data represented as physical (e.g., electronic) quantities within the computer's registers and/or memories into other data similarly represented as physical

quantities within the computer's registers and/or memories or other information storage medium that may store instructions to perform operations and/or processes.

(21) Referring now to the drawings, and more specifically to FIG. 1, what is illustrated is an exemplary automated media sheet processing system **10**, which can be used with methods herein. The system **10** is an example of a multidirectionally stepped auto-collator system that may automatically collate and transfer media sheets exiting an upstream source **50** (FIG. 3). The upstream source **50** may include a media supply **12**, such as a conventional unwinder that roll feeds media sheets in continuous sheet form into a conventional cutter **14** that cuts the continuous roll fed sheet media sheets into predetermined lengths of, e.g., about 5 inches to 8 feet, or about 8 inches to 2 feet, or about 13-14 inches. In examples, each predetermined sheet length may be cut into four separate and individual workpieces (e.g., media sheets **32** (FIG. 3)) and each workpiece may be perforated into a number (e.g., four) of parallel sections to accommodate different in-store requirements. It is understood that the predetermined sheet lengths may be cut into other numbers of separate and individual workpieces, and that each individual workpiece may be perforated into other numbers of different parallel sections. While this upstream source **50** feeds roll stock and cuts and perforates the stock, it does not include a system for automatically accumulating and collating media sheets **32** exiting the upstream source.

(22) The system **10** includes a controller **16** and at least one marking device (printing engine(s)) **18** operatively connected to the controller. The system may also include a communications port (Input/Output device **20**) operatively connected to the controller **16** and to a computerized network external to the system **10**. The Input/Output device **20** may be used for communications to and from the system **10**, as well understood by a skilled artisan.

(23) The controller **16** includes at least one processor and controls the various actions of the system **10**, as described in greater detail below. A storage medium **22** (e.g., non-transitory computer storage medium, which may be optical, magnetic, capacitor based, etc.) is readable by the controller **16** and stores instructions that the controller **16** executes to allow the system **10** to perform its various functions, such as those described herein. Thus, as shown in FIG. 1, a body housing **24** has one or more functional components that operate on power supplied from an external power source **26**, which may comprise an alternating current (AC) power source, through a power supply **28**. The power supply **28** may include a power storage element (e.g., a battery) and connects to the external power source **26**. The power supply **28** converts the power from the external power source **26** into the type of power needed by the various components of the system **10**.

(24) The system **10** has the media supply **12** providing media to a media path **30**. The media path **30** may include any combination of belts, rollers, nips, drive wheels, vacuum devices, air devices, etc. that transport the media from the media supply **12** through system **10**, as well understood by a skilled artisan. The system **10** includes a marking device (e.g., printing engine **18**) positioned along the media path **30**. The marking device prints marks on the media. Further, the cutter **14** may be positioned along the media path **30**, and the cutter divides (cuts) the media into individual workpieces or media sheets **32** (FIG. 3), such as signs.

(25) While signs are used as an example of a type of media sheet that can be processed with the system **10** herein, those ordinarily skilled in the art understand that virtually any form of workpiece that can be stacked could be used with the disclosed structures and methods, and the claims are not limited only to signs. Therefore, signs, sheets of paper, cards, pieces of plastic, etc., as well as many other items could be the media sheets processed by the systems and methods herein.

(26) A patterning device **34** may be positioned along the media path **30**. The patterning device **34** may include cutters in addition to the cutter **14** to further cut the media sheets into smaller media sheets (e.g., orthogonally to the first direction slit of the cutter **13**), such that each further cut media sheet may correspond to an individual sign. The patterning device **34** may also insert additional patterns into the media sheets **32**, such as perforations. The cutter **14** and the patterning device **34** may be combined into a single device or can be separate devices, depending upon the specific

configuration. Further, the printing engine **18**, cutter **14**, and patterning device **34** may be positioned in any order along the media path **30**, and the order shown is purely arbitrary.

(27) While not being limited to a particular configuration, system **10** may include at least one accessory functional component, such as a graphic user interface (GUI) assembly **36**, an optical scanner **38**, or other accessory functional component (such as a document handler, automatic document feeder (ADF), etc.) that operate on power supplied from the external power source **26** via the power supply **28**.

(28) A transport device **40** is additionally positioned along the media path **30**. The transport device **40** moves the media sheets **32** from the media path **128** and places the media sheets into a bidirectional collating apparatus **42**, as will be described in greater detail below. After the media sheets **32** are compiled and collated, the collated media sheets may be forwarded to additional stations, such as at least one bander **44**, for further processing and distribution as desired.

(29) As would be understood by those ordinarily skilled in the art, the system **10** shown in FIG. **1** is only one example and the systems and methods herein are equally applicable to other types of devices that may include fewer components or more components. For example, while a limited number of printing engines and media paths are illustrated in FIG. **1**, those ordinarily skilled in the art would understand that many more media paths and additional printing engines may be included within any device used with embodiments herein.

(30) FIGS. **2** and **3** depict related art media sheet processing apparatus media path sections including an accumulation and collating device **52** located directly after a transport device **40**. FIG. **4** depicts a related art accumulation and collating device **52** in side view. The accumulation and collating device **52** includes a series of ramped angled baffles **54** that accept and register media sheets **32** exiting the transport device **40** and accumulates the registered media sheets. Each angled baffle **54** is configured to allow the media sheets **32** to fall into place during an accumulation stage and then allow the media sheets to be collated into a compiled stack after all of the media sheets have been fed from transport device **40**. The angled baffles **54** may also have an aperture **94** that splits the baffles **54** into separate sections to permit transverse movement by a pusher **56**, as will be described in greater detail below.

(31) As may be seen in FIGS. **3** and **4**, the transport device **40** moves media sheets **32** to the related art accumulation and collating device **52**. The accumulation and collating device **52** may include a buffer/compiler **58** above a sequential cross-process collator **60** having the angled baffles **54**. The buffer/compiler **58** holds one or more media sheets **32** in stacks **62** above the collator **60** to allow time for the collator to actuate and reset before the next stack **62** of media sheets **32** are collated. In examples, the buffer/compiler **58** includes shelves **46** that support the stacks **62** above respective angled baffles **54**. The buffer/compiler may also include walls **48** between shelves **46** to help segregate the media sheets **32** into separate stacks **62** on the shelves.

(32) The compiling and collating process may be divided into stages. In a first stage, the media sheets **32** are output from the body housing **24** into the buffer/compiler **58** where they are compiled in stacks **62** on the shelves **46**. The media sheets **32** in the stacks **62** are in a predetermined sequential order as controlled by the controller **16**. In a second stage, also referred to as an accumulation state, the stacks **62** are dropped onto the angle baffles **54** of the collator **60**. The stacks **62** may be dropped from the shelves **46** in various ways. For example, the shelves **46** may be moveable and configured to slide out to a first position to support media sheets **32** output from the body housing **24**, and also to retract or otherwise move to a second position that allows the stacked media sheets to fall or move onto the angled baffles **54**. In other examples the stacks **62** may be urged or pushed out of the shelves **46** onto the angled baffles.

(33) The collator **60** includes the series (e.g., more than one) of the ramped angle baffles **54** and an automated pusher **56** (FIG. **4**) that moves each of the stacks **62** toward a final collated stack **66**. At the end of the accumulation stage, the pusher **56** is moved orthogonally to the media sheets **32** to push the media sheets from each ramped angled baffle **54** onto the top of media sheets in adjacent

bins of baffles **54** in succession to collate the media sheets onto a final stack platform **64** (FIG. 4). That is, during this stage, which may be considered a third stage, the pusher **64** sweeps the stacks **62** in sequential order as an interim stack **68**. Continued sweeping of the pusher across the baffles **54** to the final stack platform **64**, results in the interim stack **68** of collated media sheets **32** pushed onto the final stack platform as the final collated stack **66**. The final collated stack **66** has a predetermined number of media sheets **32** in a known order as controlled by the controller **16**. Pusher **56** may be arranged for automated pulling through openings **70** (FIG. 2) between baffles **54** towards platform **64** to unload the dropped media sheet stacks **62** from each baffle and simultaneously convey the interim stack **68** onto the platform at a stacking position thereon laterally across from the angled baffles as a collated stack **66** after all of the predetermined number of media sheets **32** have been fed from the transport device **40**.

(34) The multi-stage process allows the transport device **40** to stack the media sheets **32** in the buffer/compiler **58** for temporary holding in order to provide the time needed for previous sets of media sheets to be collated underneath. This allows the pusher **56** time to move the interim stacks **68** into a final collated stack **66** and to return to a starting (e.g., home) position prior to the buffer/compiler **58** releasing its temporary hold of the stacks **62** and dropping the next stacks down into the collator **60**.

(35) The related auto-collation apparatus discussed above thus employs a set of static angled collation baffles and a one-directional push to a single final stack platform **64**. This automated apparatus makes use of a right angle accumulation and collating device **52** which is used to compile and then sweep the stacks **62** of media sheets in only one direction with the pusher **56**. After each push of a collation the accumulation and collating device **52** resets by rewinding the pusher position to home to prepare for the next collation. Because the accumulation and collating device **52** must reset to the home position the time associated with the reset must be accommodated for in the apparatus timing. This adds significant time to the overall process. In other related systems, after each push of the pusher **56** for a collation, the accumulation and collating device **52** resets by lowering the pusher below the static angled collation baffles **54** so the pusher can traverse below the angled baffle and collating media sheets during the retract move. Then the accumulation and collating device **52** raises the pusher **56** after the traversal. However, this drop and reset to height causes additional vibration and settling issues. Even after the pusher **56** completes its traversal and reaches its home position, it is actuated into the up/push position where it must wait to settle prior to commencing the next push.

(36) Exemplary embodiments replace the fixed collation angled baffled bins, one-direction push and reset motion of related systems with a collating system of dynamic stepped bins that change height and include a bi-directional push with no reset. As discussed in greater detail below, the collating system includes the bidirectional collating apparatus **42** that overcomes any need for the above-discussed reset by collating and pushing alternatively in opposing directions, thereby reducing the time needed for continuous collation of media sheets. In examples, the collating system counts the sheets for a collation, and pushes completed sets to a corresponding bander. Once the pusher and completed sets have cleared the bins, the dynamic stepped bins move and change orientation into a stair configuration descending in a different direction, and a next collation may be collected. While not being limited to a particular theory, a second bander may be added adjacent the collation system at a location opposite or offset from the first bander so there may be a bander on different sides of the collation system. Once collected, a next completed set may be pushed to another bander. In this way the collation system may alternately push completed sets to each bander eliminating the need for a reset motion and associated timing loss. No reset is needed because the pusher ends a first push in position for the start of a next push to the opposite side/bander.

(37) FIG. 5 depicts an exemplary collating system **70** in accordance with exemplary embodiments. The collating system **70** includes the bidirectional collating apparatus **42** and may include the

buffer/compiler **58** upstream the bidirectional collating apparatus and after the body housing **24**. The bidirectional collating apparatus **42** includes a dynamically oriented baffle **72** having a plurality of dynamically stepped bins **74** (e.g., shelves, blocks, supports, escalator platforms, stairway steps) identified individually as stepped bins **82**, **84**, **86**, **88**, with each stepped bin arranged adjacent to another one of the stepped bins. The stepped bins **74** each have a top surface sized and shaped to receive and support a stack **62** of media sheet **32** thereon. In examples the top surface may be at least substantially flat and horizontal to help prevent deposited media sheets **32** from unintentionally sliding off the top surface.

(38) Each bin **74** may be configured to shift or slide dynamically relative to an adjacent bin, and to maintain its top surface horizontal orientation throughout the dynamic movement. Bins **74** may be sliding arranged next to an adjacent bin, for example, with each bin coupled to at least one lever **76** or other supporting member configured to dynamically support the bins throughout their shifting movements. The lever **76** may be a rigid rod or board pivotably attached to a fixed fulcrum **78** (e.g. rod, stem) to rotate around the fulcrum. In examples, stepped bins **74** may be attached to the lever **76** via a respective pin (not shown) extending out of each stepped bin into a longitudinal slot or channel (not shown) extending longitudinally within the lever and facing the bin. As the lever **76** pivots, the pins may slide within the lever channel to allow the dynamic stepped bins **74** to move vertically with respect to its adjacent bin(s) while maintaining its top surface orientation, as readily understood by a skilled artisan.

(39) Still referring to FIG. **5**, the bidirectional collating apparatus **42** includes an automated bidirectional pusher **80** similar to the automated pusher **56**. However, unlike the pusher **56**, the bidirectional pusher **80** is configured to push stacks **62** of media sheets **32** in multiple directions onto different platforms **64** located at opposite sides of the dynamically oriented baffle **72**. In examples, the bidirectional pusher **80** may include a substantially vertically extending bar (e.g., rod, pole, stem, shaft, support beam, handle) configured to sweep back and forth across the dynamically oriented baffle **72** without a need to rewind or reset to a home position between successive pushes. The bidirectional pusher **80** is thus an exemplary automated reciprocating pusher configured for automatic bidirectional movement in a cross process direction to the media sheets **32** deposited onto the dynamically stepped bins **74**.

(40) The bidirectional pusher **80** is shown at a first position in FIG. **5** left of the dynamically stepped bins **74**, which are shown in a neutral horizontal configuration with the stepped bins next to each other. This neutral horizontal configuration may be part of a buffer collection configuration where the bins **74** are oriented to receive stacks **62** of media sheets. Of course the buffer collection configuration is not limited to the neutral horizontal configuration and includes additional stepped bin orientations that allow deposition of media sheets onto the stepped bins in neutral, ascending or descending configurations. A stack **62** of media sheets **32** is shown deposited on each stepped bin **74** from the buffer/compiler **58**. Meanwhile, additional media sheets **32** are shown deposited on the buffer/compiler shelves **46** that may be subsequently deposited on the stepped bins **74**.

(41) FIG. **6** depicts the collating system **70** of FIG. **5** with the stepped bins **74** having a first stair configuration descending in a first direction **90** downhill from left to right in the view. While not being limited to a particular configuration, during a dynamic movement or shift, each stepped bin **74** may move vertically and change height with respect to an adjacent stepped bin. As can be seen in the stage shown in FIG. **6**, the lever **76** may be pivoted clockwise from the lever shown in FIG. **5**. This clockwise pivot shifts the stepped bins **74** left of the fulcrum **78** vertically upwards, with stepped bin **82** shifted higher relative to stepped bin **84** due to its greater distance from the fulcrum. Similarly, the clockwise lever **76** pivot shifts the stepped bins **74** that are right of the fulcrum **78** vertically downwards, with stepped bin **88** shifted lower relative to stepped bin **86** due to its greater distance from the fulcrum. The clockwise pivot results in the stepped bins **74** having a first stair configuration descending in the first direction **90**, and the top surface maintains its horizontal orientation throughout the pivot.

(42) After the lever **76** completes its clockwise pivot, or at least pivots sufficiently for the top surface of a higher stepped bin **74** to be above a height of the stack **62** in the adjacent next lower stepped bin, a first sweeping operation of the automated bidirectional pusher **80** may commence. During this first sweeping operation, the bidirectional pusher **80** sweeps the stacks **62** in sequential order along the first direction **90** onto an interim stack **68** and towards a final collated stack **66** on the right-side platform **64a**. The final collated stack **66** is formed by the automated bidirectional pusher **80** moving via a pusher drive system (not shown) that may include at least one motor (e.g., servo motor, stepper motor) that drives the bidirectional pusher back and forth across the stepped bins **74** in one of the cross process directions to stack the separate stacks **62** into a single collated stack that may be arranged in in-store planogram order, as understood by a skilled artisan.

(43) It should be noted that the collated stack **66** shown in FIGS. **5** and **6** may be moved from the right-side platform **64a** before the interim stack **68** and stacks **62** are moved to the right-side platform for further processing. Such further processing may include forwarding the collated stack **66**, for example via a rotating conveyor belt, from the right-side platform **64a** to a bander **44a** to secure the collate stack together. Similar to the ramped angled baffles **54** shown by example in FIG. **2**, the stepped bins **74** may include an aperture **94** that splits each stepped bin into a front section **92** visible in the views of FIGS. **5-8** and a back section. In examples, the bidirectional pusher **80** may sweep across the dynamically oriented baffle **72** by traversing across the aperture within the stepped bins, as readily understood by a skilled artisan.

(44) FIG. **7** depicts the collating system **70** of FIG. **5** with the stepped bins **74** back in their neutral horizontal configuration, with the automated bidirectional pusher **80** at a second position right of the dynamically stepped bins **74**, and with the collated stack **66** formed by the left-to-right sweep of the bidirectional pusher **80** on the right-side platform **64a**. In examples where FIG. **7** sequentially follows FIG. **6**, the lever **76** may be pivoted counterclockwise from the lever shown in FIG. **6**. This counterclockwise pivot shifts the stepped bins **74** left of the fulcrum **78** vertically downwards, and shifts the stepped bins **74** that are right of the fulcrum **78** vertically upwards, resulting in the stepped bins **74** shifted back to their neutral horizontal configuration, and the top surface maintaining its horizontal orientation throughout the pivot.

(45) FIG. **8** depicts the collating system **70** of FIG. **5** with the stepped bins **74** having a second stair configuration ascending in the first direction **90** uphill from left to right in the view. In examples where FIG. **8** sequentially follows FIG. **7**, the lever **76** counterclockwise pivot continues from the lever shown in FIG. **6**. Each stepped bin **74** continues vertical movement and changes height with respect to an adjacent stepped bin. This counterclockwise rotation shifts the stepped bins **74** left of the fulcrum **78** vertically downwards, with stepped bin **82** shifted lower than stepped bin **84** due to its greater distance from the fulcrum. Similarly, the counterclockwise lever **76** pivot shifts the stepped bins **74** that are right of the fulcrum **78** vertically upwards, with stepped bin **88** shifted higher than stepped bin **86** due to its greater distance from the fulcrum. The counterclockwise pivot results in the stepped bins **74** having the second stair configuration ascending in the first direction **90** opposite the second direction **96**, and the top surface maintains its horizontal orientation throughout the pivot.

(46) After the lever **76** completes its counterclockwise pivot, or at least pivots sufficiently for the top surface of a higher stepped bin **74** to be above a height of the stack **62** in the adjacent next lower stepped bin, a second sweeping operation of the automated bidirectional pusher **80** may commence. During this second sweeping operation, the bidirectional pusher **80** sweeps the stacks **62** in sequential order along the second direction **96** onto an interim stack **68** and towards a final collated stack **66** on the left-side platform **64b**. Of course the depositing of the stacks **62** onto the stepped bins **74**, the bidirectional pivoting of the lever **76**, and sweeping operations of the automated bidirectional pusher **80** may continue, for example as depicted in FIGS. **5-8**, to continue automatic collating and transferring of media sheets as desired.

(47) Bidirectional collation of the media sheets **32** may be performed after each store or job

requirement is completed and the bidirectional collation facilitates compiling of media sheets on the final stack platforms **64** in the shortest distance for reduced delay in feeding the media sheets to the multidirectionally stepped auto-collator system. Bidirectional collation of the media sheets **32** may also occur intra job, for example, upon a number (e.g., greater than 5, less than 100, between 10 and 50) of media sheets being deposited onto each stepped bin **74**. This may occur when there are too many media sheets **32** (e.g., more than 20, more than 40, more than 200) to a store that need to be swept in the cross process directions at a time. This may also occur if a store requires media sheets **32** having different lengths, where sheets having the same length may be collated at a time before another length of sheets is deposited onto the stepped bins. Thus more than one bidirectional cross process collations may occur for a store or job requirement.

(48) It should be noted that the collated stack **66** shown in FIG. **8** may be moved from the left-side platform **64b** before the interim stack **68** and stacks **62** are moved to the left-side platform for further processing. Such further processing may include forwarding the collated stack **66**, for example, automatically after the collated stack is placed on the platform **64b** from the platform to a bander **44**, **44b** via a transport unit **98** (e.g., combination of at least one conveyor belt, automated pick and place device, automated arm, slide, rollers, nips, drive wheels, vacuum devices, air devices), with the bander configured to secure the collate stack together. The bander **44b** receiving collated stacks **66** from the left-side platform **64b** may be the same bander **44** or a different bander that receives collated stacks from the right-side platform **64a**.

(49) In examples, the collated stacks **66** may be unloaded from the final stack platforms **64** for further processing and shipping as desired. Before such unloading, the collated stacks may be bound together for easier handling and transport. FIG. **9** depicts the system **10** including a plurality of banders **44** (e.g., Madison Bander by Controls Engineering LLC) integrated with the bidirectional collating apparatus **42** in top view. A first bander **44** (e.g., right-side bander **44a**) may be seen adjacent the final stack platform **64**, and a second bander (e.g., left-side bander **44b**) may be offset from their respective final stack platform **64a**, **64b** with a transport unit **98** therebetween to move collated stacks **66** to the respective bander. The banders **44** are configured to band the collated stack **66** so that all media sheets **32** in the stack are held together in one bound package. Examples are not limited to a particular type or configuration of a specific bander. In examples, the banders **44** may wrap a thin film (e.g., plastic wrap) around a collated stack **66** automatically upon placement of the stack to the bander by the transport unit **98**, as understood by a skilled artisan. In examples, the banders **44** may including a padding option, where, for example, adhesive is applied to one or more edges of a collated stack **66** to bound the stack together, as understood by a skilled artisan. The bander **44** may automatically band the compiled collated stack **66** of the media sheets **32** into a bounded bundle, for example via a sensor (e.g., camera, imager, light detector) of the bander detecting the stack (e.g., front edge, rear edge, weight, belt travel) located within the bander, and communicating the detection to the controller **16** and/or bander, as understood by a skilled artisan.

(50) The disclosed embodiments may include an exemplary method for automated accumulation, collating, transfer and packaging of media sheets **32** and sheet stacks **62** in an automated media sheet processing system **10**. FIG. **10** illustrates a flowchart of such an exemplary method. As shown in FIG. **10**, operation of the method commences at Step **S100** and proceeds to Step **S110**, where an automated media sheet processing system provides a dynamically oriented baffle having a plurality of stepped bins. Each stepped bin is slidably arranged adjacent another one of the plurality of stepped bins and has a longitudinal surface to receive media sheets deposited thereon.

(51) At Step **S110**, the system dynamically shifts the plurality of stepped bins vertically with each bin changing height with respect to an adjacent stepped bin. This dynamic shift of the stepped bins resulting in the stepped bins having a first stair configuration descending in a first direction.

Operation of the method proceeds to Step **S120**, where the stepped bins receive a first set of the media sheets from an upstream source onto the longitudinal surfaces of the stepped bins. At Step

S130, an automated reciprocating pusher configured for automatic bidirectional movement in a cross process direction to the deposited media sheets moves in a first direction to move the media sheets from the stepped bins in the first stair configuration into a first compiled collated stack at a first position. Additional media sheets may be deposited onto the stepped bins as set forth in Step **120**.

(52) Operation proceeds to Step **S140**, where the stepped bins are dynamically shifted vertically and into a different vertical orientation with respect to an adjacent stepped bin resulting in the plurality of stepped bins having a second stair configuration ascending in the first direction, or descending in a second direction opposite the first direction. Operation proceeds to Step **S150**, where the reciprocating pusher is moved in a second direction opposite the first direction to move the media sheets from the stepped bins in the second stair configuration into a second compiled collated stack at a second position distanced from the first position.

(53) Operation may cease at Step **S160** for subsequent packaging of the compiled collated stacks. In examples, subsequent packaging may include at least one banding unit adjacent the transport device that may automatically band the compiled collated stacks of the media sheets into bounded bundles. Operation may also continue by repeating back to Step **S110** for a dynamic shift of the stepped bins back to the first stair configuration and processing of additional media sheets.

(54) The exemplary depicted sequence of executable method steps represents one example of a corresponding sequence of acts for implementing the functions described in the steps. The exemplary depicted steps may be executed in any reasonable order to carry into effect the objectives of the disclosed embodiments. No particular order to the disclosed steps of the method is necessarily implied by the depiction in FIG. **10**, and the accompanying description, except where any particular method step is reasonably considered to be a necessary precondition to execution of any other method step. Individual method steps may be carried out in sequence or in parallel in simultaneous or near simultaneous timing. Additionally, not all of the depicted and described method steps need to be included in any particular scheme according to disclosure.

(55) FIG. **11** illustrates a block diagram of the controller **16** for executing instructions to automatically control the automated media sheet processing system **10** and components thereof. The exemplary controller **16** may provide input to or be a component of a controller for executing the media sheet processing method for automatically collating and transferring media sheets including controlling the media sheet accumulation, collating and transfer in a system such as that depicted in FIGS. **1, 5-9**, and described in greater detail below.

(56) The exemplary controller **16** may include an operating interface **102** by which a user may communicate with the exemplary control system. The operating interface **102** may be a locally-accessible user interface associated with the automated media sheet processing system **10**. The operating interface **102** may be configured as one or more conventional mechanism common to controllers and/or computing devices that may permit a user to input information to the exemplary controller **16**. The operating interface **102** may include, for example, a conventional keyboard, a touchscreen with “soft” buttons or with various components for use with a compatible stylus, a microphone by which a user may provide oral commands to the exemplary controller **16** to be “translated” by a voice recognition program, or other like device by which a user may communicate specific operating instructions to the exemplary controller. The operating interface **102** may be a part or a function of the graphical user interface (GUI) **36** mounted on, integral to, or associated with, the automated media sheet processing system **10** with which the exemplary controller **16** is associated.

(57) The exemplary controller **16** may include one or more local processors **104** for individually operating the exemplary controller **16** and for carrying into effect control and operating functions for automated processing of media sheets **32** for distribution to associated retail stores, including accumulating the sheets onto baffles **72**, collating the accumulated sheets into collated stacks **66**, and/or transferring the collated stacks to platforms **64** for further processing. For example, in real-

time upon receipt of media sheets **32** exiting an upstream source, processors **104** may trigger an automated bidirectional pusher **80** to shift the media sheets in collated form into a compiled collated stack **66** in store planogram order, and then move the collated stack for subsequent packaging and shipment as needed to an associated retail store. Processor(s) **104** may include at least one conventional processor or microprocessor that interprets and executes instructions to direct specific functioning of the exemplary controller **16**, and control media sheet processing with the exemplary controller.

(58) The exemplary controller **16** may include one or more data storage devices **106**, including the computer storage medium **22**. Such data storage device(s) **106** may be used to store data or operating programs to be used by the exemplary controller **16**, and specifically the processor(s) **104**. Data storage device(s) **106** may be used to store information regarding, for example, media sheet status information, media sheet accumulation information, media sheet compilation information, media sheet stack information, location information of the media sheet stacks, and other processing information with which the automated media sheet processing system **10** is associated. Stored media sheet and stack data may be devolved into data to generate a recurring, continuous or automated media sheet processing system in the manner generally described by examples herein.

(59) The data storage device(s) **106** may include a random access memory (RAM) or another type of dynamic storage device that is capable of storing updatable database information, and for separately storing instructions for execution of media sheet processing by, for example, processor(s) **104**. For example, a data storage device **106** may be coupled to the processor **104**, and may include instructions which when executed by the processor, cause the processor to direct the automated bidirectional pusher **80** to move in cross process directions to the media sheets after the media sheets have settled into the dynamically stepped bins **74** to shift the media sheets in collated form from the stepped bins into a compiled collated stack at a first or second position (e.g., stack platform **64**, **64a**, **64b**) and direct the transport unit **98** to move the compiled collated stack of media sheets from the first or second position downstream for further processing. Data storage device(s) **106** may also include a read-only memory (ROM), which may include a conventional ROM device or another type of static storage device that stores static information and instructions for processor(s) **104**. Further, the data storage device(s) **106** may be integral to the exemplary controller **16**, or may be provided external to, and in wired or wireless communication with, the exemplary controller **16**, including as cloud-based data storage components.

(60) The data storage device(s) **106** may include non-transitory machine-readable storage medium used to store the device queue manager logic persistently. While a non-transitory machine-readable storage medium is may be discussed as a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store one or more sets of instructions. The term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instruction for execution by the controller **16** and that causes the automated media sheet processing system **10** to perform any one or more of the methodologies of the present invention. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media.

(61) The exemplary controller **16** may include at least one data output/display device **108**, which may be configured as one or more conventional mechanisms that output information to a user, including, but not limited to, a display screen on a GUI **36** of the automated media sheet processing system **10** device with which the exemplary controller **16** may be associated. The data output/display device **108** may be used to indicate to a user a status of the automated media sheet processing system **10** with which the exemplary controller **16** may be associated including an operation of one or more individually controlled components at one or more of a plurality of

separate media sheet processing stations or subsystems associated with the automated media sheet processing system, including but not limited to the media supply **12**, the buffer/compiler **58**, the transport device **40**, the bidirectional collating apparatus **42**, the collaging system **70**, the transport unit **98**, and the bander(s) **44**.

(62) The exemplary controller **16** may include one or more separate external communication interfaces **112** by which the exemplary controller **16** may communicate with components that may be external to the exemplary media sheet processing system such as the media supply **12** and cutter **14**. At least one of the external communication interfaces **112** may include the input/output device **20** and be configured as an input port to support connecting an external CAD/CAM device storing modeling information for execution of the control functions in the media sheet processing operations. Any suitable data connection to provide wired or wireless communication between the exemplary controller **16** and external and/or associated components is contemplated to be encompassed by the depicted external communication interface **112**.

(63) The exemplary controller **16** may include a collate/transfer control device **114** that may be used to control a media sheet processing process including media sheet accumulation, collating and transfer. The collate/transfer control device **114** may operate as a part or a function of the processor **104** coupled to one or more of the data storage devices **106** and the automated media sheet processing system **10**, or may operate as a separate stand-alone component module or circuit in the exemplary controller **16**.

(64) All of the various components of the exemplary controller **16**, as depicted in FIG. **11**, may be connected internally, and to the automated media sheet processing system **10**, associated media sheet formation and processing devices upstream or downstream the automated media sheet processing system and/or components thereof, by one or more data/control busses **116**. These data/control busses **116** may provide wired or wireless communication between the various components of the automated media sheet processing system **10** and any associated media sheet formation and processing devices, whether all of those components are housed integrally in, or are otherwise external and connected to the automated media sheet processing system with which the exemplary controller **16** may be associated.

(65) It should be appreciated that, although depicted in FIG. **11** as an integral unit, the various disclosed elements of the exemplary controller **16** may be arranged in any combination of subsystems as individual components or combinations of components, integral to a single unit, or external to, and in wired or wireless communication with the single unit of the exemplary control system. In other words, no specific configuration as an integral unit or as a support unit is to be implied by the depiction in FIG. **11**. Further, although depicted as individual units for ease of understanding of the details provided in this disclosure regarding the exemplary controller **16**, it should be understood that the described functions of any of the individually-depicted components, and particularly each of the depicted control devices, may be undertaken, for example, by one or more processors **104** connected to, and in communication with, one or more data storage device(s) **106**.

(66) Those skilled in the art will appreciate that other embodiments of the disclosed subject matter may be practiced with many types of media sheet processing elements common to automated media sheet processing systems in many different configurations. For example, although automated media sheet processing systems and methods are shown in the discussed embodiments, the examples may apply to other types of media sheet processing systems and methods. It should be understood that these are non-limiting examples of the variations that may be undertaken according to the disclosed schemes. In other words, no particular limiting configuration is to be implied from the above description and the accompanying drawings.

(67) It will be appreciated that various of the above-disclosed and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications.

Also, various presently unforeseen or unanticipated alternatives, modifications, variations or improvements therein may be subsequently made by those skilled in the art.

Claims

1. A system for automatically collating and transferring media sheets exiting an upstream source, comprising: a dynamically oriented baffle having a plurality of stepped bins, each stepped bin slidably arranged adjacent another one of the plurality of stepped bins and having a longitudinal surface to receive a first set of the media sheets deposited thereon, the dynamically oriented baffle configured to first shift the plurality of stepped bins vertically and change height with respect to an adjacent stepped bin, the first shift of the stepped bins resulting in the plurality of stepped bins having a first stair configuration descending in a first direction; and an automated reciprocating pusher configured for movement in a cross process direction to the deposited media sheets, the automated reciprocating pusher being configured during a first operation to move the first set of media sheets in the first direction from the stepped bins in the first stair configuration into a first compiled collated stack at a first position; the dynamically oriented baffle further configured to dynamically second shift the stepped bins vertically and change vertical orientation with respect to the adjacent stepped bin, the second shift of the stepped bins resulting in the plurality of stepped bins having a second stair configuration descending in a second direction opposite the first direction, the longitudinal surfaces of the stepped bins further configured to receive a second set of the media sheets deposited thereon.
2. The system of claim 1, the automated reciprocating pusher further configured for automatic bidirectional movement in opposing cross process directions to the deposited media sheets, and during a second operation to move the second set of the media sheets in the second direction from the stepped bins in the second stair configuration into a second compiled collated stack at a second position distanced from the first position.
3. The system of claim 2, wherein the automated reciprocating pusher is further configured to move bidirectionally horizontally in the first direction and second direction absent vertical movement therebetween.
4. The system of claim 2, further comprising a transport device configured to move the first compiled collated stack of the media sheets from the first position to a third position downstream the first position in response to detecting a formation of the first compiled collated stack for subsequent packaging of the first compiled collated stack, the transport device further configured to move the second compiled collated stack of the media sheets from the second position to a fourth position downstream the second position in response to detecting a formation of the second compiled collated stack for subsequent packaging of the second compiled collated stack.
5. The system of claim 4, further comprising a first banding unit and a second banding unit, the first banding unit adjacent the transport device and configured to automatically band the first compiled collated stack of the media sheets at the third position into a bounded bundle, the second banding unit adjacent the transport device and configured to automatically band the second compiled collated stack of the media sheets at the fourth position into another bounded bundle.
6. The system of claim 2, further comprising a controller including: a processor in communication with the dynamically oriented baffle and the automated reciprocating pusher; and a storage device coupled to the processor, wherein the storage device includes instructions which, when executed by the processor, cause the processor to direct the dynamically oriented baffle to shift the plurality of stepped bins into one of the first stair configuration and the second stair configuration, and to further cause the processor to direct the automated reciprocating pusher to move in the descending direction and move the media sheets thereon in one of the first direction and the second direction.
7. The system of claim 6, further comprising a compiler positioned along the processing path between the upstream source and the dynamically oriented baffle, the compiler being operatively

connected to the controller and the dynamically oriented baffle, the compiler configured to receive the media sheets exiting the upstream source and to temporarily hold the media sheets as directed by the controller, and following one of the first operation and the second operation, the compiler further configured to release the held media sheets to the dynamically oriented baffle.

8. The system of claim 7, the compiler including a shelf configured to move from a first position where the shelf is configured to receive and hold the media sheets to a second position where the shelf is configured to release the held media sheets to the dynamically oriented baffle.

9. The system of claim 8, the shelf including a plurality of ledges, each ledge in the first position extending over a respective stepped bin.

10. The system of claim 1, further comprising a compiler positioned along the processing path between the upstream source and the dynamically oriented baffle, the compiler being operatively connected to the dynamically oriented baffle, the compiler configured to receive the media sheets exiting the upstream source and to temporarily hold the media sheets, and following the first operation, the compiler further configured to release the held media sheets to the dynamically oriented baffle.

11. The system of claim 10, the compiler including a shelf configured to move from a first position where the shelf is configured to receive and hold the media sheets to a second position where the shelf is configured to release the held media sheets to the dynamically oriented baffle.

12. The system of claim 11, the shelf including a plurality of ledges, each ledge in the first position extending over a respective stepped bin.

13. The system of claim 12, the compiler further including a plurality of guide walls each extending vertically adjacent at least one of the plurality of ledges.

14. A apparatus for automatically collating and transferring media sheets exiting an upstream source, comprising: a dynamically oriented baffle having a plurality of stepped bins, each stepped bin slidably arranged adjacent another one of the plurality of stepped bins and having a longitudinal surface to receive a first set of the media sheets deposited thereon, the dynamically oriented baffle configured to first shift the plurality of stepped bins vertically and change height with respect to an adjacent stepped bin, the first shift of the stepped bins resulting in the plurality of stepped bins having a first stair configuration descending in a first direction; a compiler positioned along the processing path between the upstream source and the dynamically oriented baffle, the compiler being operatively connected to the dynamically oriented baffle, the compiler configured to receive the media sheets exiting the upstream source and to temporarily hold the media sheets, and following one of the first operation and the second operation, the compiler further configured to release the held media sheets to the dynamically oriented baffle; an automated reciprocating pusher configured for automatic bidirectional movement in a cross process direction to the deposited media sheets, wherein, in response to detecting that the first set of the media sheets have settled into the bins for collation thereof, the automated reciprocating pusher is configured during a first operation to move the first set of the media sheets in the first direction from the stepped bins in the first stair configuration into a first compiled collated stack at a first position; the dynamically oriented baffle further configured to dynamically second shift the stepped bins vertically and change vertical orientation with respect to the adjacent stepped bin, the second shift of the stepped bins resulting in the plurality of stepped bins having a second stair configuration descending in a second direction opposite the first direction, the longitudinal surfaces of the stepped bins further configured to receive a second set of the media sheets deposited thereon; and in response to detecting that a second set of media sheets have settled into the bins for collation thereof, the automated reciprocating pusher is further configured during a second operation to move the second set of the media sheets in the second direction from the stepped bins in the second stair configuration into a second compiled collated stack at a second position distanced from the first position, and the automated reciprocating pusher is further configured to move bidirectionally horizontally in the first direction and second direction without vertical movement therebetween.

15. A method for automatically collating and transferring media sheets exiting an upstream source, comprising: providing a dynamically oriented baffle having a plurality of stepped bins, each stepped bin slidably arranged adjacent another one of the plurality of stepped bins and having a longitudinal surface to receive media sheets deposited thereon, the dynamically oriented baffle configured to first shift the plurality of stepped bins vertically and change height with respect to an adjacent stepped bin, the first shift of the stepped bins resulting in the plurality of stepped bins having a first stair configuration descending in a first direction; receiving a first set of the media sheets from the upstream source onto the longitudinal surfaces of the plurality of stepped bins; providing an automated reciprocating pusher configured for movement in a cross process direction to the deposited media sheets; during a first operation, moving the automated reciprocating pusher in a first direction to move the first set of the media sheets from the stepped bins in the first stair configuration into a first compiled collated stack at a first position; and dynamically shifting the stepped bins vertically and into a different vertical orientation with respect to the adjacent stepped bin resulting in the plurality of stepped bins having a second stair configuration descending in a second direction opposite the first direction, the plurality of stepped bins further configured to receive a second set of media sheets deposited thereon.

16. The method of claim 15, further comprising, during a second operation, moving the automated reciprocating pusher in a second direction opposite the first direction to move the media sheets from the stepped bins in the second stair configuration into a second compiled collated stack at a second position distanced from the first position, the automated reciprocating pusher further configured for automatic bidirectional movement in opposing cross process directions to the deposited media sheets.

17. The method of claim 16, further comprising moving the automated reciprocating pusher bidirectionally horizontally in the first direction during the first operation and in the second direction during the second operation absent vertical movement therebetween.

18. The method of claim 16, further comprising moving the first compiled collated stack of the media sheets from the first position to a third position downstream the first position with a transport device for subsequent packaging of the first compiled collated stack, and moving the second compiled collated stack of the media sheets from the second position to a fourth position downstream the second position with the transport device for subsequent packaging of the second compiled collated stack.

19. The method of claim 18, further comprising banding the first compiled collated stack of the media sheets at the third position into a bounded bundle with a first banding unit adjacent the transport device, and banding the second compiled collated stack of the media sheets at the fourth position into another bounded bundle with a second banding unit adjacent the transport device.

20. The method of claim 16, further comprising providing a compiler positioned along the processing path between the upstream source and the dynamically oriented baffle, the compiler being operatively connected to the dynamically oriented baffle, receiving receive the media sheets exiting the upstream source and temporarily holding the media sheets with the compiler, and following one of the first operation and the second operation, releasing the held media sheets to the dynamically oriented baffle via the compiler.
